Open Instruction-Tuned LLMs: A Reproducible Evaluation across Tasks

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ABSTRACT

Natural Language Processing (NLP) for lesser-resourced languages faces persistent challenges, including limited datasets, inherited biases from high-resource languages, and the need for domain-specific solutions. This study addresses these gaps for Modern Greek through three key contributions. We evaluate the performance of open-source large language models (LLMs) on seven core NLP tasks for which Greek data are available and properly licensed, revealing task-specific strengths, weaknesses, and parity in their performance. These contributions provide a roadmap to advance NLP in Greek, bridging gaps in model evaluation, task innovation, and real-world impact.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; *Language resources*; • **Information systems** → Data analytics.

KEYWORDS

Greek NLP, benchmarking, large language models, reproducibility, evaluation

1 INTRODUCTION

Natural Language Processing (NLP) has advanced substantially with the rise of large language models (LLMs), whose development depends on immense volumes of digital data [7]. Although multilingual NLP has benefited from these advances, progress for lesser-resourced languages continues to lag behind that of well-supported languages. As a result, NLP for many of the world's languages often relies on methods, resources, and evaluation practices developed for high-resource settings, inheriting assumptions, biases, and other characteristics that may not align with the properties of less supported languages [3].

This gap is compounded by the scarcity of open-access, high-quality language resources. In this context, high-quality resources refer to native, human-authored corpora, as opposed to synthetic or machine-translated data. Although synthetic and translated resources are often used to bridge data gaps, they can introduce "translationese", where the text inherits syntactic and stylistic constraints from the source language [19], and may lack the lexical diversity and morphological nuance of original language use. Such resources are needed both for pre-training foundation models and for building gold-standard benchmarks for downstream tasks. Comprehensive surveys of NLP resources for lesser-resourced languages therefore do more than document digital readiness: they can also guide future research, reveal reusable datasets, and support the construction of new benchmarks. Our work focuses on one such language, Modern Greek.[1]

In this work, we ask the following research question: *which properly licensed Greek datasets can be used to train and/or assess LLMs on downstream NLP tasks?* By addressing this question, this work makes three key contributions:

- Based on a recent Greek NLP survey [3], we extract publicly available Greek datasets that are (a) open-access and (b) permit derivatives, and aggregate them into a unified collection that is easy to recompile and use.[2]
- We classify the collected datasets according to their usability for benchmarking or pre-training.
- We benchmark open LLMs with manageable parameter counts and Greek-language capabilities across multiple NLP tasks, revealing task-specific strengths and weaknesses.

2 RELATED WORK

2.1 Greek NLP Resources and Models

Modern Greek NLP has a long tradition of task-specific resources and systems, including work on grammar checking, legal text processing, named entity recognition, sentiment analysis, and syntactic annotation [1, 4, 13, 28, 34]. More recent surveys have shown, however, that these resources remain unevenly distributed across tasks and domains, and that availability, licensing, and reusability are recurring barriers for reproducible experimentation [3, 26, 30]. This motivates our focus on datasets that are not only relevant to downstream Greek NLP, but also accessible and legally reusable.

Transformer-based models have also expanded the practical toolkit for Greek. Greek-specific pretrained models such as GreekBERT and GreekBART improved representation learning and sequence-to-sequence modelling for the language [11, 21], while recent open LLMs for Greek, such as Meltemi, indicate growing interest in language-specialised generative models [41]. In parallel, multilingual and open instruction-tuned LLMs such as Llama, Gemma, Aya, Qwen, and Falcon provide broader coverage and are increasingly used through local serving frameworks [2, 9, 16, 39, 45]. Our work complements these model-development efforts by evaluating manageable open models under a shared Greek benchmark protocol.

2.2 Benchmarking and Evaluation

Multi-task benchmarks such as GLUE and SuperGLUE established shared evaluation as a central mechanism for comparing general-purpose language understanding systems [42, 43]. Similar efforts have since appeared for individual languages, domains, and multilingual settings, including legal and lower-resource evaluation suites [8, 40, 44]. For foundation models and LLMs, broad evaluation frameworks such as HELM and the LM Evaluation Harness further emphasize reproducibility, task coverage, and transparent reporting [6, 14, 22].

[1] In our study, the term "Greek" refers exclusively to Modern Greek (ISO 639-1: el), excluding previous stages of the language's history.

[2] The code is available at: anonymised.url

Despite this progress, Greek LLM evaluation remains fragmented across datasets, task definitions, metrics, and execution environments. Existing Greek studies often target individual tasks or model families, making it difficult to compare open instruction-tuned LLMs across heterogeneous downstream settings. Our benchmark therefore prioritizes a reproducible evaluation pipeline over a new modelling method: it brings together properly licensed Greek datasets, standardizes task execution and reporting, and produces outputs that can be inspected, reproduced, and reused in subsequent work.

3 THE DATA: FROM COLLECTION TO DISTILLATION

Based on a systematic literature review covering 142 research NLP studies for Greek from 2012 to 2023 [3], we extracted and analysed information on 94 publicly available Greek datasets. Most of these resources were monolingual (84%), but bilingual and multilingual datasets were also present (i.e., with Greek being one of the languages). The most frequent domains were politics and news (20%) while the most frequent task was sentiment analysis (33%).

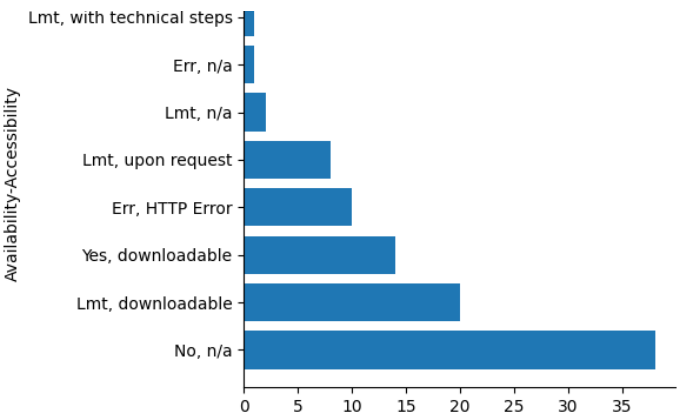

Figure 1: Availability and accessibility of collected datasets (tuple per bar). Availability is classified to Yes, Lmt (no license or otherwise constrained), Err (hindered by an error), No (not available via a URL). Accessibility is classified to downloadable, HTTP error, upon request, n/a (missing resource); i.e., the outcome we observed when accessing the resource.

3.1 Availability, Popularity and Licensing

Figure 1 shows the availability status of the collected datasets. The most common status is that **the datasets are not available**, meaning no information is provided by the authors of the respective study regarding their availability. The second most frequent category concerns datasets reported as downloadable, but **actually not publicly available** due to restrictions, such as missing license or subscription limitations. This limitation contradicts the principles and benefits of open access.

Studies with publicly available datasets, serving as a reference point, tend to attract more citations. This is shown by the higher third and fourth quartile of the green box in Figure 2. Furthermore, studies attracting citations while presenting data with limited [17]

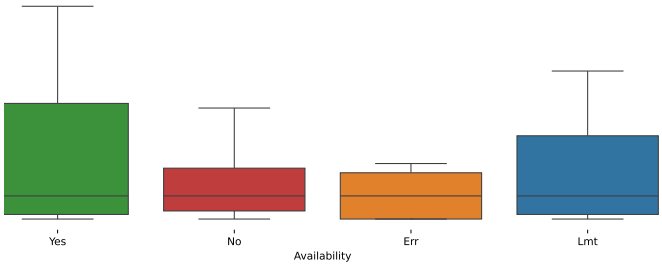

Figure 2: Whisker and box plots with the number of citations of studies per availability type.

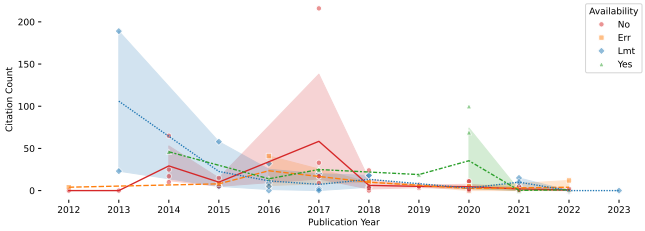

Figure 3: Citation counts per year of publication of studies developing datasets, based on the availability classified as yes, no, limited (lmt), or erroneous (err).

or no availability [15] belong to the past. The study of Juola and Stamatatos [17] is such an example, shown in Figure 3 with a blue peak in 2013, concerning a shared task on authorship identification (hence, the citations) whose data lack a license (hence, the limitation).

3.2 AIR Distillation

We collected accessible, interoperable and re-usable (AIR)[3] datasets pooled from a recent survey [3]. We focused on the 14 Greek datasets that meet our criteria of being licensed, accessible, machine-actionable, and with no hidden costs. We excluded one [12] that was licensed under CC BY-NC-ND 4.0, which does not allow derivatives and thus limits reusability. We also excluded one [46] that shared tweets not via rehydration (i.e., only IDs shared), but as full texts.[4] We also excluded a dataset of 100 sentences [20], which is too small for our study. And we excluded the dataset of open-access e-books [5], because it comprises OCR artifacts.[5] Given these exclusions, we end up with ten datasets, presented in Table 1, which cover a variety of domains.

[3]To comply with the FAIR principles, datasets must have a unique and persistent digital identifier (commonly a Persistent Identifier/PID). A PID is a specific combination of numbers and letters that ensures datasets can be uniquely and sustainably accessed as digital objects over time. Merely providing a URL for accessibility does not make a dataset findable, as URLs can become obsolete or break if not properly maintained.
[4]Although it is possible that a company gained access to the texts (e.g., via firehose), releasing them with another license, a user should still be able to delete their content from all platforms. If not, then this dataset is not compliant with the rules. If so, then the dataset is not stable, harming reproducibility of experiments.
[5]Although typos and errors are not prohibited for experiments, we observe that the author names and the title was often injected in the main text, leading to data leaking for potential benchmarks.

Table 1: Greek datasets with their annotation type (AT) defined as manual (M), automatic (A), curated (C; metadata provided by the distributor), hybrid (H; manual and automatic), user-generated (U; from user-edits, not curated) or no (N). We include also the host, size, size unit, domain and license of each dataset.

AUTHORS	AT	HOST	SIZE	SIZE UNIT	DOMAIN	LICENSE
Papantoniou et al. [29]	A	Zenodo	18,615	DOC	EVENTS	CC BY 3.0
Koniaris et al. [18]	*C*	Hugging Face	8,395	DOC	LEGAL	CC BY-NC 4.0
Rizou et al. [37]	*M*	msensis.com	2,176	SENT	ADMIN	CC-BY-NC-SA 4.0
Dritsa et al. [10]	A	Zenodo	1.28M	DOC	POLITIC	CC BY 4.0
Papaloukas et al. [28]	*C*	Hugging Face	47,563	DOC	LEGAL	CC BY 4.0
Korre et al. [20]	*M*	GitHub	227	SENT	ESSAYS	MIT
Bartziokas et al. [4]	*H*	GitHub	21,153	SENT	NEWS	CC-BY-NC-SA 4.0
Prokopidis and Piperidis [36]	N	nlp.ilsp.gr	101,857	DOC	WEB	CC-BY-NC-SA 4.0
Prokopidis and Papageorgiou [34]	*H*	GitHub	2,521	SENT	NEWS, POLITIC	CC BY-NC-SA 3.0
Prokopidis et al. [35]	*U*	nlp.ilsp.gr	17,018	DOC PAIR	NEWS	CC BY 4.0

Table 2: AIR datasets for RAG (not in the benchmark) with min, max, mean length in characters and size in documents.

	MIN	MAX	AVG	SIZE (#)
Papantoniou et al. [29]	34	1,484	172.9	18,615
Prokopidis and Piperidis [36]	36	67,963	2,430	101,857
Dritsa et al. [10]	1	291,949	900	1,280,927

3.3 AIR Resources Suitable for RAG

Table 1 comprises three datasets with no ground truth or whose ground truth is automatically extracted (A/N; i.e., of doubted quality). Although excluded from our benchmark, these datasets are still possibly useful for retrieval-augmented generation (RAG) and other purposes (e.g., pre-training). These concern 1.28M political speeches, spanning from July 1989 to July 2020 [10]; Greek Wikipedia event pages with named entity recognition (NER) and linking; 101,857 (cleaned) websites with open content [36], including online archives of Greek newspapers from 2003 to 2020, and the Greek part of the W2C corpus [24].

3.3.1 Dataset statistics. As shown in Table 2, the text length (measured in characters) greatly varies between the datasets. The dataset with the shortest texts on average is that of Papantoniou et al. [29] and that with the lengthiest is that of Prokopidis and Piperidis [36]. The dataset size (counted in number of files) also varies, starting from ten thousands [29], to one hundred thousands [36], to more than a million [10]. The aggregated text totals up to 1.4 billion characters and 211 million tokens (using white space split).

3.3.2 Linguistic distance. We investigate the linguistic associations between these datasets, using statistical language modelling (LMing) as a proxy. For each dataset, we trained one statistical character-level LM on the first 100 characters of 1,000 randomly selected texts per dataset.[6] Then, we computed the Bits per Characters (BPC) across 500 randomly selected texts per dataset, reporting the mean. The result is displayed as a heatmap in Figure 4. High scores (in red) reflecting a greater linguistic distance between the dataset the LM is trained on and the one BPC is computed on. Low scores (in blue) reflect a low linguistic distance, with the lowest scores taking place in the diagonal, when predicting unseen text taken from the same dataset (self-distance). Based on this heatmap, we find that the dataset of Prokopidis and Piperidis [36] shows a higher self-distance compared to the rest while the datasets of Dritsa et al. [10] and Papantoniou et al. [29] are far away from each other, especially when the LM is trained on the former and predicting the latter.

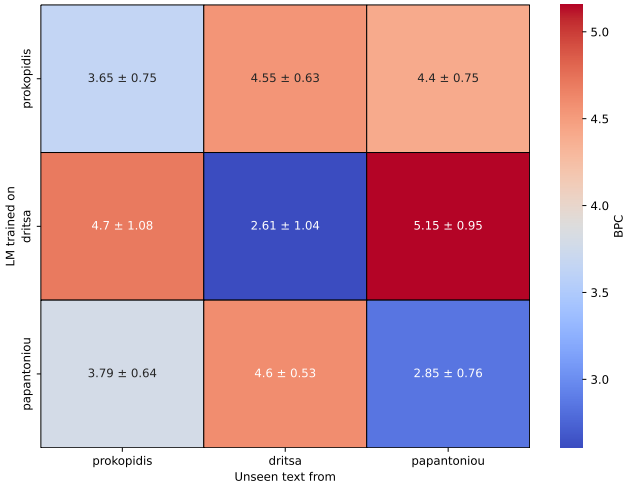

Figure 4: Heatmap of BPC measured on a sample per RAG dataset (average across the sample; shown horizontally) per trained LM (vertically); greater distance in warm colours.

[6] Character-level statistical LMing is an efficient way to capture stylistic features, employed in authorship analysis [31].

3.4 Resources Suitable for Downstream Tasks

The datasets used as benchmarks for downstream Greek NLP tasks are summarized in Table 3, along with key statistics and their associated supervised tasks. These resources cover three main categories: classification (Intent, Legal), sequence labeling (POS tagging, NER), and text-to-text generation (Grammatical Error Correction, Machine Translation, Summarization). Next, we present each one.

3.4.1 Intent Classification. Student queries, addressed to two university help desks, were manually annotated with six intent labels and three entity types [37]. This dataset is balanced across intent classes, with relatively short texts (average length: 58.6 characters). We use it to evaluate intent classification.

3.4.2 Legal Text Classification. A multi-level legal classification dataset is derived from the "Permanent Greek Legislation Code – Raptarchis" [28]. It includes hierarchical annotations of 47 volumes, 347 chapters, and 1,685 subjects. The dataset is highly imbalanced and exhibits large variation in document length (from 17 to approx. 1 million characters). The largest class comprises 34 examples (6.8%; science and arts) and the median has 10 examples. The smallest class has two examples (0.4%; media and tourism). We use it for volume classification.

3.4.3 POS Tagging. The Greek Universal Dependencies treebank was part of a project offering standardized treebanks with consistent annotations across languages [27]. This dataset [34] includes syntactic dependencies, POS tags, morphological features, and lemmas. It consists of 2,521 manually validated sentences split into train (1,622), development (403), and test (456) sets.

3.4.4 NER. The Greek NER dataset [4] comprises two annotation schemes: a coarse-grained set of 4 entity types (similar to CONLL-2003 [38]) and a fine-grained set of 18 types (similar to OntoNotes [33]). The dataset was developed during a Google Summer of Code project in 2018,[7], where initial automatic annotation was followed by manual curation. The text length varies from 2 to 2,210 characters (Table 3). Both annotation levels exhibit heavy imbalance, with the four-tag level having an imbalance ratio (minor:major) of 0.02 and the 18-tag level showing near-zero imbalance. We assess NER at the less imbalanced four-entity-tag level.

3.4.5 Grammatical Error Correction (GEC). The Greek Native Corpus (GNC) [20] comprises (227) student-written sentences, annotated by human experts for (18) error types, related to grammar, spelling, punctuation, and morphology. Each sentence may contain zero, one, or multiple grammatical errors. We use this dataset to assess grammatical error correction.

3.4.6 Machine Translation (MT). Bilingual corpora (756 language pairs) exist from content available in Global Voices, where volunteers translate news stories in 41 languages [35]. The 3,629 Greek documents are translated into 40 languages. However, not every document is translated into all languages, resulting in 17,018 bilingual document pairs involving Greek. We use the translations to English, Japanese, and Farsi, representing high-, moderately-, and low-supported language tiers [3].

[7] https://github.com/eellak/gsoc2018-spacy

Table 3: Dataset statistics for downstream Greek NLP tasks. We report test set size, number of classes (#C), average text length (in characters), and class imbalance ratio (CIR; smallest-to-largest class proportion) where applicable. Datasets split by us are marked with †.

Task	Test	Unit	#Cl.	Avg Char.	CIR
Intent Cl. [37]	436†	sent	6	58.6	1.00
Legal Text Cl. [28]	500	text	47	3673.3	0.07
POS Tagging [34]	456	sent	16	138.2	0.03
NER [4]	500†	sent	18	180.9	0.02
GEC [20]	175	sent	-	483.0	-
MT [35]	1,500†	doc	3	125.1	-
Summarization [18]	300	text	-	18,771.3	-

3.4.7 Summarization. A legal corpus of 8,395 court decisions is compiled from the Greek Supreme Court (Areios Pagos),[8] including the full texts, their summaries, case categories, and associated tags [18]. This dataset exhibits substantial length variability, with 18,541 characters on average and 18,351 st. dev. (min: 2,571; max: 303,538). We use it to assess summarization.

4 BENCHMARK SETUP

4.1 Tasks and Models

Our benchmark covers seven task families: GEC; intent classification; legal classification; machine translation; NER; POS tagging; summarization. We evaluate five open-source and instruction-tuned models, which could be served locally through Ollama; namely:

- `qwen2.5:7b-instruct` [45];[9]
- `llama3.1:8b-instruct` [16];[10]
- `aya-expanse:8b` [9];[11]
- `gemma2:9b` [39];[12]
- `falcon3:7b-instruct` [2].[13]

4.2 Deterministic Capped-Test Protocol

The full test sets are very uneven in size, with legal classification and machine translation being especially large. To keep evaluation tractable while preserving reproducibility, we use a deterministic capped-test profile that always takes the first instances in dataset order (as suggested by the dataset providers) instead of random samples. The resulting capped evaluation (Table 3) contains:

- GEC: 175 examples
- intent classification: 436 examples
- legal classification: 500 examples
- machine translation: 500 English pairs, 500 Persian pairs, and 500 Japanese pairs
- NER: 500 examples
- POS: 456 examples
- summarization: 300 examples

[8] https://www.areiospagos.gr/
[9] https://huggingface.co/Qwen/Qwen2.5-7B-Instruct
[10] https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct
[11] https://huggingface.co/CohereLabs/aya-expanse-8b
[12] https://huggingface.co/google/gemma-2-9b
[13] https://huggingface.co/tiiuae/Falcon3-7B-Instruct

This design keeps the smaller tasks intact while reducing the runtime burden of the largest tasks. It also supports exact reruns because the selected subset is deterministic. For machine translation, the benchmark now uses a shared data-wrapper path rather than a separate task-local split implementation, which eliminates an earlier source of inconsistency in language-specific counts.

4.3 Metrics

We report task-appropriate primary metrics. For GEC, the primary metric is GLEU [25]; we also report exact match and edit-distance diagnostics (WER/CER) against reference and source text. For intent classification, legal classification, NER, and POS tagging, macro-F1 is the primary comparison metric, with accuracy as a supplementary metric; for NER we additionally report entity-level F1. For machine translation, BLEU is the primary metric, while chrF [32] and WER/CER are reported as supplementary diagnostics. For summarization, BERTScore-F1 [47] is the primary metric, with ROUGE-1/2/L [23] and WER/CER reported as supplementary overlap and string-distance diagnostics. Latency is reported as average per-instance generation time in seconds.

To compare models across heterogeneous metrics, we additionally compute a direction-aware normalized quality score per benchmark segment. Within each segment, the best model receives the highest normalized value and the worst model the lowest. These segment-normalized scores are then averaged to obtain an overall ranking. The main paper focuses on the primary metrics used for ranking, while the appendix reports additional prompt and winner-margin material.

5 EXPERIMENTAL RESULTS

We analyse the experimental results by focusing on the best model per task, but also touching on the relationship between performance and efficiency. Table 4 summarizes the best model in each benchmark segment. Figure 5 visualizes normalized quality across segments, while Figure 6 shows the trade-off between quality and inference speed. Furthermore, we provide an analytical table per task (keeping corresponding figures in the Appendix).

5.1 Patterns across models

5.1.1 The best model. The strongest overall model is `gemma2:9b`, ranking first on the normalized aggregate with an average quality score of 80.8 and winning intent classification, legal text classification, NER, POS, and two of the three machine translation segments.

5.1.2 The runner up complements the best. `llama3.1:8b-instruct` ranks second overall with an average normalized quality score of 62.5. It wins GEC and remains highly competitive in intent classification and summarization, but it trails `gemma2:9b` on legal text classification and the aggregate ranking.

5.1.3 Competitive in cross lingual tasks. Third is `qwen2.5:7b-instruct`, which is especially competitive on Persian machine translation and wins that segment. `aya-expanse:8b` ranks fourth overall, but remains a strong specialist in summarization and is the runner-up on legal text classification and English translation.

5.1.4 The weakest model is not uniformly weak. The weakest model is `falcon3:7b-instruct`. This is often among the fastest models even when it lags in quality, which reinforces the value of reporting latency alongside task scores.

Table 4: Best-performing model per benchmark segment on the deterministic capped evaluation run (GEC=175, Intent=436, Legal=500, MT(ENG=500, FAS=500, JPN=500), NER=500, POS=456, Summarization=300), using representative metrics per task.

Task Segment	Winner	Primary Metric	Value
GEC	Llama 3.1 8B Instruct	GLEU	0.796
Intent	Gemma2 9B	Macro-F1	0.737
Legal	Gemma2 9B	Macro-F1	0.126
MT-ENG	Gemma2 9B	BLEU	35.073
MT-FAS	Qwen2.5 7B Instruct	BLEU	4.183
MT-JPN	Gemma2 9B	BLEU	6.227
NER	Gemma2 9B	Macro-F1	0.122
POS	Gemma2 9B	Macro-F1	0.486
Summ.	Aya Expanse 8B	BERTScore-F1	0.519

Table 5: Per-model results for GEC.

Model	GLEU	Exact Match	Latency (s)
Llama 3.1 8B Instruct	0.796	0.120	0.822
Gemma2 9B	0.784	0.137	1.117
Qwen2.5 7B Instruct	0.751	0.069	1.651
Aya Expanse 8B	0.655	0.057	0.772
Falcon3 7B Instruct	0.617	0.006	1.892

Table 6: Per-model results for Intent.

Model	Accuracy	Macro-F1	Latency (s)
Gemma2 9B	0.862	0.737	0.284
Llama 3.1 8B Instruct	0.720	0.707	0.224
Qwen2.5 7B Instruct	0.752	0.645	0.204
Aya Expanse 8B	0.716	0.563	0.285
Falcon3 7B Instruct	0.578	0.488	0.188

Table 7: Per-model results for Legal.

Model	Accuracy	Macro-F1	Latency (s)
Gemma2 9B	0.346	0.126	1.349
Aya Expanse 8B	0.292	0.098	1.070
Llama 3.1 8B Instruct	0.182	0.076	0.914
Qwen2.5 7B Instruct	0.202	0.051	1.417
Falcon3 7B Instruct	0.122	0.047	1.014

5.2 Patterns across tasks

5.2.1 Text classification. Legal text classification remains a difficult benchmark component, although the refreshed coarse-grained

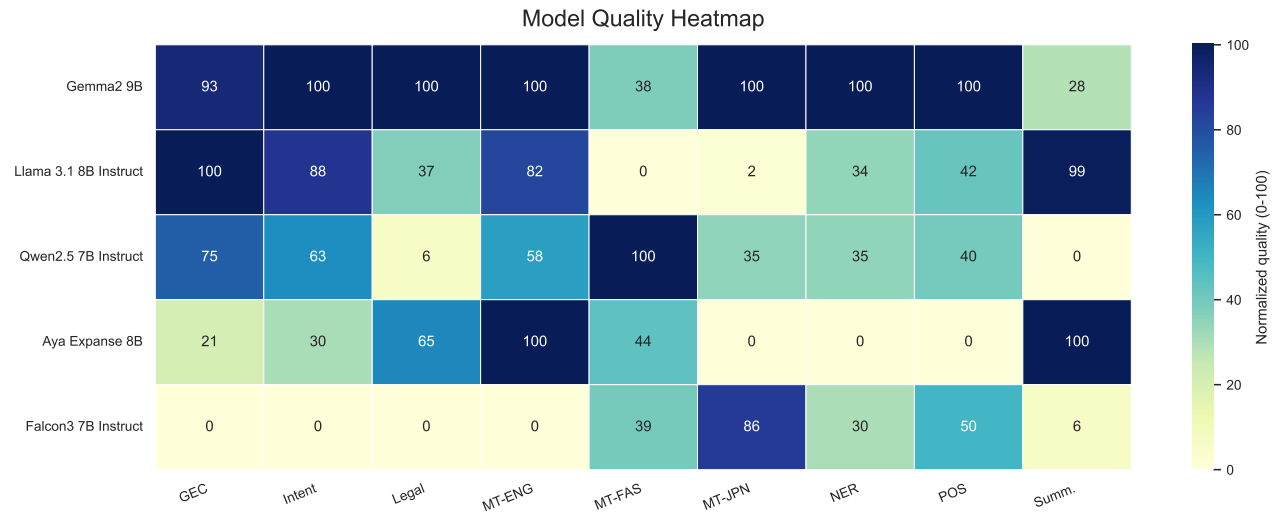

Figure 5: Normalized quality heatmap across the nine benchmark segments. Scores are normalized within each segment so that higher is always better, even when the underlying metric is lower-is-better such as WER.

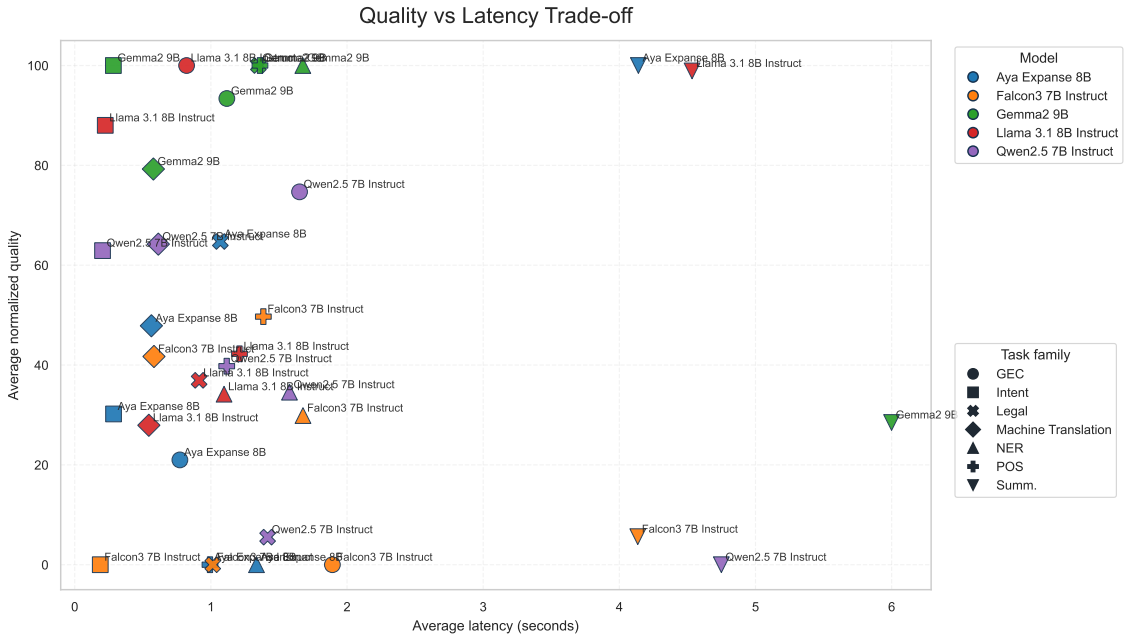

Figure 6: Quality-latency trade-off across models and task families. Each point corresponds to one model-task pair after averaging within the task family.

setup raises the best macro-F1 to 0.126 with `gemma2:9b`.[14] This suggests that the task is still demanding under zero-shot prompting and may benefit from better label descriptions, exemplars, or task-specific fine-tuning. By contrast, intent classification is comparatively mature, with `gemma2:9b` reaching 0.737 macro-F1.

5.2.2 Translation. The capped profile evaluates 500 examples per target language, which is enough to separate models consistently while keeping runtime manageable. Using BLEU as the primary metric, `gemma2:9b` leads English and Japanese translation, while `qwen2.5:7b-instruct` is first on Persian (see Appendix metrics for chrF and WER/CER diagnostics).

[14]We replaced opaque identifiers with coarse-grained volume labels.

Table 8: Per-model results for Machine Translation.

Lang	Model	WER	BLEU	chrF	Latency (s)
ENG	Gemma2 9B	57.915	35.073	61.781	0.569
ENG	Aya Expanse 8B	56.359	35.037	61.591	0.531
ENG	Llama 3.1 8B Instruct	64.237	32.576	60.045	0.441
ENG	Qwen2.5 7B Instruct	66.484	29.280	56.907	0.439
ENG	Falcon3 7B Instruct	77.902	21.355	49.132	0.431
JPN	Gemma2 9B	109.538	6.227	16.726	0.629
JPN	Falcon3 7B Instruct	115.930	5.665	8.895	0.900
FAS	Qwen2.5 7B Instruct	116.337	4.183	18.325	0.845
JPN	Qwen2.5 7B Instruct	168.901	3.656	12.553	0.558
JPN	Llama 3.1 8B Instruct	168.112	2.358	14.845	0.588
JPN	Aya Expanse 8B	109.067	2.279	16.533	0.610
FAS	Aya Expanse 8B	109.222	2.146	3.408	0.548
FAS	Falcon3 7B Instruct	114.451	1.984	3.124	0.416
FAS	Gemma2 9B	110.071	1.926	2.906	0.536
FAS	Llama 3.1 8B Instruct	343.614	0.555	2.058	0.604

Table 9: Per-model results for NER.

Model	Accuracy	Macro-F1	Entity F1	Latency (s)
Gemma2 9B	0.796	0.122	0.043	1.675
Qwen2.5 7B Instruct	0.689	0.095	0.045	1.577
Llama 3.1 8B Instruct	0.812	0.094	0.049	1.097
Falcon3 7B Instruct	0.804	0.093	0.026	1.677
Aya Expanse 8B	0.735	0.080	0.010	1.335

Table 10: Per-model results for POS.

Model	Accuracy	Macro-F1	Latency (s)
Gemma2 9B	0.603	0.486	1.361
Falcon3 7B Instruct	0.438	0.332	1.385
Llama 3.1 8B Instruct	0.390	0.309	1.210
Qwen2.5 7B Instruct	0.354	0.302	1.116
Aya Expanse 8B	0.229	0.180	0.993

Table 11: Per-model results for Summ..

Model	WER	ROUGE-1	ROUGE-L	BERTScore-F1	Latency (s)
Aya Expanse 8B	208.784	0.064	0.063	0.519	4.140
Llama 3.1 8B Instruct	173.585	0.081	0.076	0.517	4.534
Gemma2 9B	271.162	0.028	0.027	0.407	5.999
Falcon3 7B Instruct	235.214	0.009	0.009	0.371	4.135
Qwen2.5 7B Instruct	327.113	0.016	0.014	0.362	4.749

5.2.3 POS and NER. `gemma2:9b` leads on macro-F1 in POS and NER, but `llama3.1:8b-instruct` is better on Accuracy in NER (not POS), suggesting differences in minority-entity handling.

5.2.4 Summarization. The winner is `aya-expanse:8b`, but only by a very small margin from `llama3.1:8b-instruct` (0.519 vs. 0.517), which is also better in WER. This suggests that the top two models are effectively near-tied semantically.

Table 12: Overall ranking of the evaluated models across the nine benchmark segments in the deterministic capped benchmark run. Normalized quality is averaged over segment-level, direction-aware scores.

Model	Rank	Avg. Quality	Avg. Latency (s)	Segment Wins
Gemma2 9B	1	80.8	1.50	6
Llama 3.1 8B Instruct	2	62.5	1.16	1
Qwen2.5 7B Instruct	3	47.3	1.40	1
Aya Expanse 8B	4	46.7	1.14	1
Falcon3 7B Instruct	5	10.2	1.34	0

6 DISCUSSION

The benchmark is useful because it combines multiple Greek NLP settings in a single reproducible workflow, but the current evaluation still has limitations. The capped protocol is intentionally pragmatic rather than exhaustive. It is well suited for iterative comparison and paper-scale reporting, but it is not a substitute for full test-set evaluation when final archival numbers are required. Some metric choices are also provisional. In particular, summarization and machine translation would benefit from stronger semantic metrics and human assessment, while legal classification would benefit from richer label semantics. Even so, the appendix-level supplemental metrics already improve interpretability relative to a WER-only or macro-F1-only summary.

Despite these caveats, the benchmark already exposes meaningful structure in model behavior. Some models offer strong general performance across multiple tasks, while others are better understood as specialists for translation or summarization. This kind of cross-task perspective is hard to obtain from isolated task runs and is exactly where a unified benchmark becomes valuable.

6.1 Automatic Dataset Mining and URL Recovery

To support the continuous expansion of Greek NLP benchmarks, we implement an automated version of the search and screening protocol of Bakagianni et al. [3] over recent literature (2024–2026) retrieved from the *ACL Anthology* and *Semantic Scholar* ($n = 118$).[15] The objective is to provide a proof of concept for an *evergreen* pipeline for identifying new Greek NLP datasets.

The screening process was applied in two stages. First, non–peer-reviewed publications were removed using rule-based filtering. The remaining exclusion criteria were then handled with a zero-shot LLM classifier. This process yielded 55 in-scope papers, of which 48 were identified as likely containing dataset references based on indications in the abstract that a dataset was introduced or released.

For these papers, dataset URLs were recovered automatically through a retrieval pipeline: article text was extracted, URLs were identified with regular expressions, and *Llama 3 70B* selected the most plausible dataset link. This produced 18 candidate dataset URLs, 13 of which were verified as reachable.

A manual audit on a dataset-annotated subset ($n = 18$) showed high precision and moderate recall (precision = 1.00, recall = 0.69), indicating a conservative but reliable approach. Finally, titles and

[15]The implementation is available in our GitHub repository, enabling reuse and continuous application of the protocol for future dataset discovery.

abstracts were classified into `canonical_task` categories using *Llama 3 70B*, enabling task-level organization of newly identified resources, mainly in text classification and machine translation.

Overall, the results indicate that automatic mining can effectively complement manual curation and support the systematic, continuous enrichment of Greek NLP resources.

7 REPRODUCIBILITY AND ARTIFACT STRUCTURE

The repository includes the benchmark runner, combined CSV outputs, HTML task visualizations, report generation scripts, and publication-ready assets. The capped run used in this draft is stored under the repository's completed-run hierarchy, and the article-facing tables and vector figures are generated into a dedicated `article_assets` subdirectory. This structure allows the manuscript to be regenerated directly from benchmark outputs without manual copy-paste into the paper.

8 CONCLUSION

We presented a reproducible Greek NLP benchmark draft for open instruction-tuned LLMs and used it to compare five models across seven task families and nine benchmark segments. Under a deterministic capped-test protocol, `gemma2:9b` ranks first overall, while `llama3.1:8b-instruct`, `qwen2.5:7b-instruct`, and `aya-expanse:8b` emerge as specialists on GEC, Persian translation, and summarization respectively. The current pipeline is already useful for comparative evaluation, and it also provides a clean path from benchmark execution to publication assets.

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A WINNER MARGINS BY SEGMENT

Figure 7 presents the gap between the best and second-best model for each benchmark segment. The largest winner margins occur on NER and POS tagging.

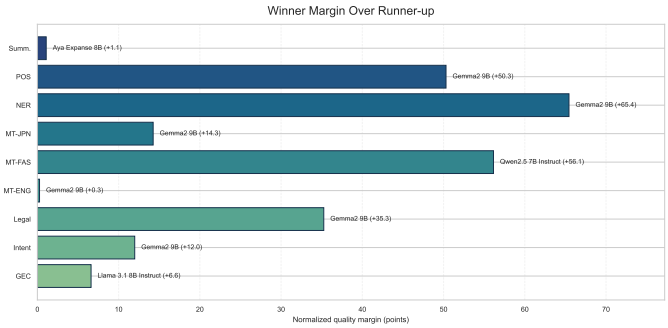

Figure 7: Normalized quality margin between the winning model and the runner-up for each benchmark segment. Large margins indicate clearer task-level separation between the top two models.

B PROMPT TEMPLATES

This appendix lists the system prompts and user-side task templates used in the benchmark. Concrete examples are instantiated by replacing angle-bracket placeholders with task-specific content at runtime.

B.1 GEC

System prompt.

```
You are a careful grammatical error correction system for Modern Gree
Correct the user's sentence while preserving its meaning.
Return only the corrected Greek sentence and nothing else.
```

Prompt template.

```
Diorthose to parakato elliniko keimeno grammatika kai orthografika.
Kratise to idio noima.

Keimeno: <ORIGINAL_TEXT>
Diorthomeno keimeno:
```

B.2 Intent Classification

System prompt.

```
You are a Greek intent classification system.
Return only the intent label and nothing else.
```

Prompt template.

```
Given a text, provide the intent of the text. Only the intent should be returned.
Here is the list of possible intents: <LABEL_SPACE>

Text: <TEXT>
Intent:
```

B.3 Legal Classification

System prompt.

```
You are a Greek legal text classification system.
Return only one label from the provided label set and nothing else.
```

Prompt template.

```
Given the following Greek legal text, return only its coarse-grained legal volume label.
Choose exactly one label from: <LABEL_SPACE>

Text: <TEXT>
Label:
```

B.4 Machine Translation

System prompt.

```
You are a translation system.
Translate the user's text accurately.
Return only the translation and nothing else.
```

Prompt template.

```
Given a text in Greek, translate it to <TARGET_LANG>.
Only the translation should be returned.

Text: <SOURCE_TEXT>
Translation:
```

B.5 Named Entity Recognition

System prompt.

```
You are a Greek named entity recognition system.
Return only a Python-style list of tags and nothing else.
```

Prompt template.

```
You are given a tokenized Greek sentence. Return only a Python-style list of
NER tags, one tag per token.
Use only these labels: <NER_LABEL_LIST>

Tokens: <TOKEN_LIST>
Tags:
```

B.6 POS Tagging

System prompt.

```
You are a Greek POS tagging system.
Return only a Python-style list of tags and nothing else.
```

Prompt template.

```
You are given a tokenized Greek sentence. Return only a Python-style list of
POS tags, one tag per token.
Use only these labels: <POS_LABEL_LIST>

Tokens: <TOKEN_LIST>
Tags:
```

B.7 Summarization

System prompt.

```
You are a Greek summarization system.
Produce a concise summary in Greek and return only the summary.
```

Prompt template.

```
Given a Greek legal text, provide its summary also in Greek.
Only the summary should be returned.

Text: <DOCUMENT_TEXT>
Summary:
```

C PROMPT TEMPLATES

This appendix lists the system prompts and user prompt templates used in the benchmark. Angle-bracket placeholders denote task-specific values replaced at runtime.

Table 13: Prompt templates used for benchmark evaluation.

Task	System Prompt	User Prompt Template
GEC	You are a careful grammatical error correction system for Modern Greek. Correct the user's sentence while preserving its meaning. Return only the corrected Greek sentence and nothing else.	Diorthose to parakato elliniko keimeno grammatika kai orthografika. Kratise to idio noima. Keimeno: <ORIGINAL_TEXT> Diorthomeno keimeno:
Intent Classification	You are a Greek intent classification system. Return only the intent label and nothing else.	Given a text, provide the intent of the text. Only the intent should be returned. Here is the list of possible intents: <LABEL_SPACE> Text: <TEXT> Intent:
Legal Classification	You are a Greek legal text classification system. Return only one label from the provided label set and nothing else.	Given the following Greek legal text, return only its coarse-grained legal volume label. Choose exactly one label from: <LABEL_SPACE> Text: <TEXT> Label:
Machine Translation	You are a translation system. Translate the user's text accurately. Return only the translation and nothing else.	Given a text in Greek, translate it to <TARGET_LANG>. Only the translation should be returned. Text: <SOURCE_TEXT> Translation:
Named Entity Recognition	You are a Greek named entity recognition system. Return only a Python-style list of tags and nothing else.	You are given a tokenized Greek sentence. Return only a Python-style list of NER tags, one tag per token. Use only these labels: <NER_LABEL_LIST> Tokens: <TOKEN_LIST> Tags:
POS Tagging	You are a Greek POS tagging system. Return only a Python-style list of tags and nothing else.	You are given a tokenized Greek sentence. Return only a Python-style list of POS tags, one tag per token. Use only these labels: <POS_LABEL_LIST> Tokens: <TOKEN_LIST> Tags:
Summarization	You are a Greek summarization system. Produce a concise summary in Greek and return only the summary.	Given a Greek legal text, provide its summary also in Greek. Only the summary should be returned. Text: <DOCUMENT_TEXT> Summary: